Quantum transport test

For a one-dimensional wave packet, the normalized state is

psi(x,t) = 1 / (2*pi*sigma_t^2)^(1/4) * exp[-(x-x0-vt)^2/(4*sigma_t^2) + i/hbar*(p0*x - p0^2*t/(2*m))]

sigma_t = sigma_0 * sqrt(1 + (hbar*t/(2*m*sigma_0^2))^2)

The measured error for three discretizations is shown below.

Grid size N	128	256	512
Relative error	1.7 x 10^-3	4.2 x 10^-4	1.0 x 10^-4
Runtime (s)	0.08	0.31	1.24

The expected convergence rate is O(N^-2).

Task: reconstruct this page faithfully as Markdown and LaTeX; preserve equations and table values.